Cardiovascular System (CARD) · NPLEX-P1-CARD-18

CV-18-C Ischemic Heart Disease

Work each activity on blank paper with the notes chapter closed, then open the answer key and mark what you missed. Do the two brain dumps for a competency on different days rather than back to back.

Work each activity without your notes first. Rate your confidence before you check. Then read the notes, write down what you missed, and set a date to come back to it.

Competency tracker

Competency	BD1 done	BD2 done	Confidence 1 to 4	Next review date
NPLEX-P1-CARD-18-03-C01				
NPLEX-P1-CARD-18-03-C02				
NPLEX-P1-CARD-18-03-C03				
NPLEX-P1-CARD-18-03-C04				
NPLEX-P1-CARD-18-03-C05				
NPLEX-P1-CARD-18-03-C06				
NPLEX-P1-CARD-18-03-C07				

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-03-C01-BD1

Myocardial oxygen supply and demand from memory

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-C Ischemic Heart Disease Competency: NPLEX-P1-CARD-18-03-C01 Activity: NPLEX-P1-CARD-18-03-C01-BD1 Date: _____

Without looking at your notes, divide a page into supply and demand. On the supply side write the resting coronary flow in mL/min and as a share of cardiac output, how much of the oxygen the myocardium extracts at rest, how many fold flow can rise in heavy exercise, and the local signals that dilate the resistance arterioles. Write the formula for the driving pressure that perfuses the left ventricle and name the phase of the cardiac cycle in which that perfusion happens. On the demand side list the three determinants and write the relationship that sets wall stress from pressure, radius, and wall thickness. Then write three sentences explaining why the subendocardium becomes ischemic before the subepicardium.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-03-C01-BD2

Four patients with chest pain and clean coronary arteries

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-C Ischemic Heart Disease Competency: NPLEX-P1-CARD-18-03-C01 Activity: NPLEX-P1-CARD-18-03-C01-BD2 Date: _____

Four patients have exertional chest pain, and coronary angiography shows no significant stenosis in any of them. Patient 1 has severe aortic stenosis with left ventricular hypertrophy. Patient 2 has a hemoglobin of 6.5 g/dL (reference range 12.0 to 16.0 g/dL). Patient 3 has untreated hyperthyroidism with a pulse of 136/min. Patient 4 used cocaine two hours before the pain started. For each patient write which determinants of oxygen balance are disturbed, whether the problem is on the supply side, the demand side, or both, and the mechanism in two or three sentences. Then write which layer of the ventricular wall becomes ischemic in patients 1 to 3 and why, and say which patient can instead develop ischemia through the whole thickness of the wall and what makes that possible.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Prediction exercise NPLEX-P1-CARD-18-03-C01-PR01

Predict what happens when demand rises on a fixed stenosis

A man has a fixed 75% stenosis of the left anterior descending artery and has been free of symptoms at rest. He develops pneumonia with a temperature of 39.2 degrees Celsius (102.6 degrees Fahrenheit), a pulse of 134/min, a blood pressure of 86/50 mm Hg, and a hemoglobin of 7.2 g/dL (reference range 13.5 to 17.5 g/dL). For each item below write the direction of the change and the mechanism in one line.

What changes	Up, down, or no change	Why, in one or two lines
What happens to the oxygen content of each mL of coronary blood		
What happens to the time available in each cardiac cycle for left ventricular perfusion		
What happens to the pressure gradient that drives subendocardial flow		
What happens to the oxygen used per minute by the myocardium		
What happens to the arteriolar reserve distal to the stenosis		
Which layer of the wall becomes ischemic first and what the ST segment does in the leads facing it		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-03-C02-BD1

Stable angina and the moving ST segment

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-C Ischemic Heart Disease Competency: NPLEX-P1-CARD-18-03-C02 Activity: NPLEX-P1-CARD-18-03-C02-BD1 Date: _____

Write out what angina pectoris is and why the pain is felt in the chest, left arm, neck, and jaw, naming the chemicals released by ischemic myocytes, the nerves they travel in, and the spinal levels involved. Then write the stenosis severity that produces exertional angina, state how resistance changes when the radius of a vessel is halved and name the relationship that gives that result, and explain in three or four sentences why flow is normal at rest but cannot rise with exertion. Explain why nitroglycerin relieves the pain, naming the second messenger and which vessels dilate most. Finish with the reason the ST segment is normally flat, and write why subendocardial ischemia depresses it while ischemia through the whole wall thickness elevates it, and which of the two localizes the artery.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-03-C02-BD2

Sort three episodes of chest pain by mechanism

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-C Ischemic Heart Disease Competency: NPLEX-P1-CARD-18-03-C02 Activity: NPLEX-P1-CARD-18-03-C02-BD2 Date: _____

Three patients come to the emergency department. Patient 1 has 5 minutes of chest pressure that comes on after two blocks of walking, eases with rest, and is accompanied by ST depression in V4 to V6 during the pain. Patient 2 has 25 minutes of chest pressure at rest that is more severe than any previous episode, with ST depression and T wave inversion, and serial troponin stays below the upper reference limit. Patient 3 is a 38-year-old smoker with chest pain at 3 in the morning, transient ST elevation in II, III, and aVF during the pain, and no stenosis on angiography. For each patient name the condition, the lesion in the artery, the layer of the wall that is ischemic, and the expected troponin result, and explain in two or three sentences how the lesion produces that ECG pattern. Then write what single change would make patient 2's diagnosis a non-ST-elevation myocardial infarction, and name the agents other than atherosclerosis that can provoke patient 3's mechanism.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-18-03-C02-RT01

Rebuild the ischemic cascade

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-C Ischemic Heart Disease Competency: NPLEX-P1-CARD-18-03-C02 Activity: NPLEX-P1-CARD-18-03-C02-RT01 Date: _____

Rebuild the six steps of the ischemic cascade from the moment demand exceeds supply to the moment the patient feels pain, one step per line and in order. Beside the step where relaxation slows, write which pump needs ATP and what that pump is doing. Beside the step where the ST segment moves, write what is different about the membrane potential of ischemic cells and what that difference generates between ischemic and normal muscle. Then write one sentence on what the position of pain in this order means for a patient with diabetic autonomic neuropathy, and one sentence on what reverses the whole cascade when no myocytes have died.

Rebuild from memory: The ischemic cascade

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-18-03-C02-CP01

Stable, unstable, and Prinzmetal angina

Fill in every cell from memory, then check it against the notes and correct what you missed.

Feature	Stable angina	Unstable angina	Prinzmetal angina
Lesion in the artery			
Pattern and timing of the pain			
Layer of the ventricular wall that is ischemic			
ECG during the pain			
Troponin result			
Typical patient and main provoking factors			
Main risk if it is left alone			

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-03-C03-BD1

From a ruptured cap to a dead myocyte

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-C Ischemic Heart Disease Competency: NPLEX-P1-CARD-18-03-C03 Activity: NPLEX-P1-CARD-18-03-C03-BD1 Date: _____

Write the sequence that runs from rupture of a fibrous cap to transmural necrosis, one step per line, naming the material exposed in the plaque, the protein that binds platelet glycoprotein Ib, the two agonists that recruit more platelets, the receptor that binds fibrinogen, and the factor that drives thrombin and fibrin. Give the time from occlusion to loss of contraction, the time to irreversible injury, and the time for necrosis to reach the epicardium. Then write what is happening inside the myocyte at each stage: which pathway makes ATP once oxygen is gone and what accumulates, which pump fails and what enters the cell, which ion floods in and which enzymes it activates, and the two structural changes that mark the point of no return. Finish by naming the three proteins that leak into blood and the time of day and the two triggers that make rupture more likely.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-03-C03-BD2

Two infarcts and two reperfused hearts

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-C Ischemic Heart Disease Competency: NPLEX-P1-CARD-18-03-C03 Activity: NPLEX-P1-CARD-18-03-C03-BD2 Date: _____

A man has a complete occlusion of the left anterior descending artery that stays occluded for 8 hours. A woman has 30 minutes of partial occlusion of the same artery that then opens. For each one write how far the necrosis extends through the wall, what the ST segment does, whether Q waves are expected, what happens to troponin, and which label fits the electrical picture. Then write which of the two can cause pericarditis and rupture of the free wall and why the other rarely does. Next, consider two hearts reperfused at 3 hours. Write what the infarct looks like on gross examination, name the change seen in the hypercontracted myocytes and what makes them hypercontract, and explain how reactive oxygen species and calcium opening the mitochondrial permeability transition pore kill cells that might have lived. Finish with the one situation in which a subendocardial infarct runs right around the ventricle instead of following one artery, and say why.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-18-03-C03-RT01

Rebuild the path from plaque rupture to transmural infarction

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-C Ischemic Heart Disease Competency: NPLEX-P1-CARD-18-03-C03 Activity: NPLEX-P1-CARD-18-03-C03-RT01 Date: _____

Draw a cross section of a coronary artery with a disrupted plaque, then rebuild the seven-step sequence from cap rupture to transmural necrosis beside it, numbering each step on the drawing. Mark on the drawing where von Willebrand factor acts, where the glycoprotein IIb/IIIa receptor acts, and where tissue factor comes from. Underneath, draw the ventricular wall as three layers and shade how far the necrosis has spread at 30 minutes, at 2 hours, and at 6 hours, writing the layer that dies first beside the drawing. Then write the two things that can stop the wavefront before it reaches the epicardium.

Rebuild from memory: From plaque rupture to transmural infarction

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-03-C04-BD1

Build the coronary territory table from memory

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-C Ischemic Heart Disease Competency: NPLEX-P1-CARD-18-03-C04 Activity: NPLEX-P1-CARD-18-03-C04-BD1 Date: _____

On blank paper build a table of infarct territories with one row each for anterior, lateral, inferior, right ventricular, and posterior. Use the columns artery, leads with ST elevation, and associated findings, and fill every cell. Write beside the table which wall and which part of the septum each of the three main arteries supplies, which artery usually supplies the sinoatrial and atrioventricular nodes, where the posterior descending artery arises and in what share of hearts, and which artery is occluded most often with its approximate frequency. Then write what reciprocal ST depression tells you and the condition it helps you rule out. Finish by writing why a pathologic Q wave appears over dead muscle and give the width and height criteria from the notes.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-03-C04-BD2

Read the biomarkers across four days

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-C Ischemic Heart Disease Competency: NPLEX-P1-CARD-18-03-C04 Activity: NPLEX-P1-CARD-18-03-C04-BD2 Date: _____

Draw one set of axes with time from 0 to 10 days on the horizontal axis and plasma concentration on the vertical axis, and sketch the curves for troponin, the MB isoenzyme of creatine kinase, and myoglobin after an infarction. Label each curve with its first rise, its peak, and the time it returns to normal. Write why troponin stays high so much longer than the MB isoenzyme, naming the two intracellular pools of troponin and where the MB isoenzyme sits in the cell. Then a patient who had an anterior infarct four days ago has chest pain again. Write which marker can show new necrosis and which cannot, and why. Write what you would conclude if that marker is normal and the pain is sharp, pleuritic, and worse lying flat. Finish by listing five conditions other than infarction that raise troponin and write the two things the definition of infarction requires beyond a single high value.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-18-03-C04-Q01

Hypotension with clear lungs after an inferior infarction

A 61-year-old man comes to the emergency department because of 2 hours of chest pressure and nausea. Temperature is 36.8 degrees Celsius (98.2 degrees Fahrenheit), pulse is 44/min, respirations are 18/min, and blood pressure is 82/54 mm Hg. Jugular venous pressure is increased and the lungs are clear on auscultation. The electrocardiogram shows ST elevation in leads II, III, and aVF with ST depression in I and aVL, and additional ST elevation in lead V4R. Which of the following best explains the hypotension with clear lung fields in this patient?

A. Acute mitral regurgitation from rupture of the posteromedial papillary muscle

B. Blood in the pericardial sac restricting ventricular filling

C. Right ventricular infarction limiting delivery of blood to the left heart

D. Loss of more than 40% of left ventricular muscle with pulmonary congestion

Why did you pick it?

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-03-C05-BD1

Build the healing timeline from memory

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-C Ischemic Heart Disease Competency: NPLEX-P1-CARD-18-03-C05 Activity: NPLEX-P1-CARD-18-03-C05-BD1 Date: _____

On blank paper build the morphologic timeline of an infarct with one row for each of these intervals: 0 to 30 minutes, 30 minutes to 4 hours, 4 to 24 hours, 1 to 3 days, 3 to 7 days, 7 to 10 days, 10 to 14 days, 2 to 8 weeks, and over 2 months. Use the columns gross appearance, light microscopy, and the complication linked to that stage, and fill every cell. Write beside the table why dead myocytes keep their outlines for several days, naming what ischemia and acidosis do to structural proteins and to lytic enzymes, and why adult myocardium is replaced by scar instead of new muscle. Then write the direction healing travels through an infarct and what that means for the center of a large one. Finish with three early details: why wavy fibers appear at the border within hours, what triphenyltetrazolium chloride stains and after how long an infarct looks pale with it, and when neutrophils arrive and what draws them.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-03-C05-BD2

Date four infarcts from what the tissue shows

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-C Ischemic Heart Disease Competency: NPLEX-P1-CARD-18-03-C05 Activity: NPLEX-P1-CARD-18-03-C05-BD2 Date: _____

Four hearts come to autopsy. Heart 1 shows a yellow-tan softened center with a hyperemic border, and microscopy shows disintegrating myocytes with macrophages at the edge. Heart 2 shows no gross change, and microscopy shows complete coagulative necrosis with loss of nuclei and dense neutrophils. Heart 3 shows a depressed red-gray border, and microscopy shows new vessels and collagen among fibroblasts. Heart 4 shows a firm white area, and microscopy shows dense collagen with few cells. For each heart give the age of the infarct as a range, name the stage of healing, and write the complication that heart is most at risk of, with the mechanism in one sentence. Then write which of the four hearts has the weakest wall and explain from the balance between removal of dead muscle and deposition of collagen why that stage is the weakest. Finish by writing what a pale unstained zone on a tetrazolium-stained heart tells you when light microscopy of the same tissue looks normal.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-03-C06-BD1

Complications laid out on a time line

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-C Ischemic Heart Disease Competency: NPLEX-P1-CARD-18-03-C06 Activity: NPLEX-P1-CARD-18-03-C06-BD1 Date: _____

Draw a horizontal time line from the moment of occlusion out to several months and place every complication from the notes on it at the time it occurs: ventricular fibrillation, bradycardia and atrioventricular block, pump failure and cardiogenic shock, right ventricular infarction, fibrinous pericarditis, free wall rupture, septal rupture, papillary muscle rupture, mural thrombus, true aneurysm, Dressler syndrome, and scar-related ventricular tachycardia. Beside the arrhythmia entries write the three electrical changes in ischemic myocytes that set up reentry and the four other factors that add to the instability. Write how much of the left ventricle must be lost for cardiogenic shock, and write why an inferior infarct slows the heart while bundle branch block from an anterior septal infarct carries a worse outlook. Then rebuild the six-step cycle of cardiogenic shock after a large infarction in order.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-03-C06-BD2

Four patients deteriorate after an infarction

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-C Ischemic Heart Disease Competency: NPLEX-P1-CARD-18-03-C06 Activity: NPLEX-P1-CARD-18-03-C06-BD2 Date: _____

Four patients deteriorate in hospital. Patient 1, four days after an inferior infarct, develops sudden pulmonary edema and hypotension with a soft apical murmur. Patient 2, five days after an anterior infarct, develops a new loud holosystolic murmur at the left lower sternal border and the oxygen saturation of blood sampled from the right ventricle is higher than expected. Patient 3, four days after an anterolateral infarct, suddenly loses her pulse with distended neck veins and muffled heart sounds while the monitor still shows organized electrical activity. Patient 4, two days after a transmural infarct, has sharp pain that is worse lying flat and a scratching sound over the precordium. For each patient name the complication, the anatomy or tissue that failed, and why that failure happens at that point in the healing sequence. For patient 1 write why one papillary muscle ruptures more often than the other, naming the arteries involved. Then write what separates a pseudoaneurysm from a true aneurysm in wall composition, neck shape, timing, and risk of rupture, and write how Dressler syndrome differs in timing and mechanism from patient 4's problem.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Clinical application NPLEX-P1-CARD-18-03-C06-CA01

Sudden breathlessness on the fifth day after an inferior infarction

A 67-year-old woman is in hospital on the fifth day after an inferior myocardial infarction. The admission electrocardiogram showed ST elevation in leads II, III, and aVF, and troponin peaked on the first day and has been falling since. While sitting up for breakfast she becomes abruptly breathless. Temperature is 37.1 degrees Celsius (98.8 degrees Fahrenheit), pulse is 118/min, respirations are 32/min, and blood pressure is 78/46 mm Hg. There are crackles throughout both lung fields, the jugular venous pressure is not raised, and a soft apical systolic murmur is heard. The electrocardiogram shows sinus tachycardia with the inferior Q waves present since admission and no new ST elevation. A repeat troponin is higher than the value from the previous morning but lower than the first day peak.

1. Recognize. Name the structural complication and the valve lesion it has produced, and name the exact structure that has failed.

2. Explain. Explain why this structure fails on the fifth day, using the healing sequence and the strength of the tissue at that point.

3. Differentiate. Write how this differs from free wall rupture, from septal rupture, and from reinfarction, naming the finding in this patient that separates each one.

4. Investigate. Write which findings you would look for to confirm the lesion and what each would show, including why the murmur can be soft rather than loud.

5. Decide. State which papillary muscle has failed and why that one rather than the other.

6. Justify. Justify the answer from the blood supply of the two papillary muscles and from the artery that supplied this infarct, and explain the rising troponin without new ST elevation.

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-18-03-C06-Q02

Timing of free wall rupture

A 58-year-old woman who had her first myocardial infarction 5 days ago suddenly loses consciousness. The monitor shows organized electrical complexes at a rate of 96/min but no pulse is palpable. The jugular veins are distended, the heart sounds are muffled, and she dies despite resuscitation. Autopsy shows a transmural anterolateral infarct with a tear through the free wall and 400 mL of blood in the pericardial sac; the heart is not hypertrophied and no collateral vessels are seen. Which of the following best explains why the wall tore on the fifth day?

A. Macrophages have removed the necrotic myocytes and collagen deposition has not yet strengthened the wall

B. Neutrophil elastase is digesting the fibrous cap of the culprit coronary plaque

C. Granulation tissue has matured into a thin fibrous scar that bulges during systole

D. Dense collagen in the healed infarct conducts slowly and sets up a reentrant circuit

Why did you pick it?

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-03-C07-BD1

The chronically ischemic heart and its remodeling cycle

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-C Ischemic Heart Disease Competency: NPLEX-P1-CARD-18-03-C07 Activity: NPLEX-P1-CARD-18-03-C07-BD1 Date: _____

Write what chronic ischemic heart disease is, what usually causes it, and which form of heart failure it is one of the commonest causes of. Then describe the heart as it would appear at autopsy: size and weight, the state of the left ventricular chamber and wall, what the scars look like and where they sit, the state of the coronary arteries, and the two findings on the endocardium. Write the three microscopic findings, including the change in subendocardial myocytes that marks chronic sublethal ischemia. Write the one feature that separates this from other dilated cardiomyopathies and what the coronary arteries look like in those. Then rebuild the six-step remodeling cycle in order, and write beside it which hormone systems are activated, which two of their mediators promote fibrosis, how sarcomeres are added in eccentric hypertrophy, and what happens to the shape of the chamber.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-03-C07-BD2

Work the cost of dilation and sort recoverable muscle from scar

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-C Ischemic Heart Disease Competency: NPLEX-P1-CARD-18-03-C07 Activity: NPLEX-P1-CARD-18-03-C07-BD2 Date: _____

Take a ventricle generating 120 mm Hg with a radius of 2.5 cm and a wall 1.0 cm thick, and a remodeled ventricle generating the same 120 mm Hg with a radius of 3.5 cm and an average wall thinned to 0.8 cm. Write the relationship for wall stress, calculate both values, show your working, give the ratio, and write in two sentences what that ratio means for oxygen demand in a heart whose arteries are already narrowed and why hypertrophy only partly helps. Next, a patient with an ejection fraction of 25% has three regions that contract poorly. Write how hibernating myocardium, stunned myocardium, and scar differ in viability, in the flow they are receiving, and in whether function can return, and say which one restoring flow cannot help. Then write how dilation produces mitral regurgitation without any disease of the valve itself and what that regurgitation adds to the ventricle. Finish by writing why slow conduction at scar borders produces monomorphic ventricular tachycardia, and name the two findings on examination and the one on the electrocardiogram that follow from a dilated ventricle with old infarcts.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____